

TUSHAR JAIN

Co-Founder · Full Stack Developer · Embedded Systems Engineer · MVP Builder

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SUMMARY

Engineering student at BMS College of Engineering (CSBS, 2024–2028) with a track record of shipping real-world products — from a health-tech wearable pitched at JIC Accelerator 7.0, to a metro app with 100+ Play Store downloads in 10 days, to a B2B SaaS reservation system. Works across the full stack: React/Next.js frontends, Supabase backends, ESP32/Arduino embedded firmware, and BLE hardware. Driven by a philosophy of utility over ornamentation every build solves a real user problem.

EXPERIENCE

Co-Founder & Tech Lead -- Pulse-Predict (Vital Health Tech)

Aug 2025 – Jan 2026

Jaipur, India · Pitched at JIC Accelerator 7.0 · tusharjain.in/work/vital-health-tech

- Identified a critical gap in India's geriatric care market — fall detection & vitals monitoring devices priced above ₹10,000 excluded the mass market. Conceptualized a sub-₹2,000 solution targeting 140M+ elderly Indians.
- Sole builder: architected the ESP32 + MPU6050 hardware stack, wrote BLE/Serial firmware, built the companion mobile app, and launched a public website from scratch.
- Solved critical false-positive challenges in fall detection through iterative motion threshold tuning during user testing, achieving a reliable alert system.
- Validated business model against 3 direct competitors at JIC Accelerator 7.0; secured industry feedback that shaped the product roadmap and sub-₹2,000 pricing strategy.
- Designed architecture for future ABDM (National Digital Health Mission) integration, enabling scale into India's official digital health infrastructure.
- Tech Stack: ESP32, MPU6050, MAX30102 (SpO2), BLE/Bluetooth, C++, Embedded Firmware, Android (APK Build & Publish).

Web Development Intern - 3Skill Training

- Completed a structured web development internship covering frontend fundamentals, responsive design, and deployment workflows. [View Certificate](#)

PROJECTS

1. Jaipur Ride (Metro) Public Utility Smart Transport Android

[Live Demo](#) · [Google Play Store](#) · [GitHub](#) · [Project Page](#)

- Centralized platform for Jaipur Metro route planning, timings, and fare calculation solving the fragmented transport info landscape that relied on WhatsApp rumors and scattered Google searches.
- Achieved 100+ downloads within 10 days of Google Play Store launch, demonstrating immediate market-fit and real-world utility.
- Designed a custom lightweight data architecture to handle transit route nodes and varying schedules without the overhead of full GIS databases keeping the app fast on low-connectivity networks.
- Built mobile-first responsive UI focused on accessibility and speed for commuters on the move.
- Tech Stack: React.js, JavaScript, HTML/CSS, Vercel.

2. Namma Ride (Bengaluru Metro) Public Utility UX Multilingual

[Project Page](#) · tusharjain.in/projects

- Bengaluru Metro journey planner with full support for English, Hindi, and Kannada ensuring accessibility for the city's diverse demographic.
- Built with vanilla JavaScript to maintain an ultra-lightweight footprint for users on weak cellular networks; complex journey-planner logic with multilingual state transitions.
- Coming soon to Android Google Play Store; building on learnings from Jaipur Ride's successful Play Store launch.
- Tech Stack: HTML, CSS, JavaScript (Vanilla).

3. BookMySlot B2B SaaS Operations Real time

[Live Demo](#) · [GitHub](#) · [Project Page](#)

- Verticalized B2B SaaS reservation system targeting Indian small businesses (restaurants as BookMyDine, clinics as BookMyClinic) that still operate on paper, WhatsApp, and phone calls.
- Designed a full system not just one app with three synchronized touchpoints: customer-facing reservation website, staff/waiter live sync app, and owner admin panel with full operational visibility.
- Used Supabase real-time database to sync state instantly across all three interfaces; Google Sheets API for lightweight owner reporting.
- Focused on Indian business reality: ultra-low cost, mobile-first, and simple enough for non-technical staff to operate without training.
- Tech Stack: HTML/CSS/JS, Supabase (Auth + Real-time DB), Vercel, Google Sheets API.

4. Air Guitar Embedded Systems Audio Synthesis Hardware

[GitHub](#) · [LinkedIn Post](#) · [Project Page](#)

- Gesture-controlled virtual guitar that synthesizes string sounds in real-time using the Karplus-Strong algorithm no pre-recorded samples, making it feel like a real physical instrument.
- MPU6050 IMU tracks wrist tilt (string selection) and flicking acceleration (strumming velocity); data streamed over Serial at 115200 baud to a Python audio engine.
- Achieved sub-10ms latency by implementing thread-safe audio callbacks; applied delta-based calibration to mitigate Serial data jitter for a responsive 'strum' feel.
- Velocity-sensitive logic adjusts synthesized sound based on acceleration magnitude, creating dynamic playing expression.
- Tech Stack: Python, Arduino, MPU6050 (IMU), NumPy, Karplus-Strong Synthesis, Serial Communication (115200 baud).

5. Indigo In-Flight - Air-Stream Systems Offline First Web

[GitHub](#) · [Project Page](#)

- Offline-first in-flight entertainment web system inspired by how airlines stream content without internet built to explore air-gapped local network constraints.
- Designed fully offline-capable frontend with no external CDN dependencies; assets load reliably over a local server without internet access.
- Architected for multi-device delivery (phones, tablets) over a local network simulating real airline seatback entertainment delivery infrastructure.
- Key insight: offline systems require fundamentally different UX assumption's reliability and predictability matter more than feature richness.
- Tech Stack: HTML, CSS, JavaScript, Offline-first architecture.

6. Split-Payment System FinTech System Design Prototype

[GitHub](#) · [LinkedIn Post](#) · [Project Page](#)

- Concept-level fintech system enabling a single purchase to be split across multiple payment sources simultaneously (e.g., UPI + Credit Card).
- Designed transaction atomicity logic: if any one payment source fails, the entire transaction is safely rolled back no partial charges or duplicate deductions.
- Integrated Razorpay test mode to simulate real payment gateway interactions; explored edge cases including retries, timeout failures, and partial authorization scenarios.
- Tech Stack: JavaScript, Node.js, Razorpay (Test Mode Integration).

7. NotesCSBS Productivity Academic Platform React

[Live Demo](#) · [Project Page](#)

- Centralized academic resource hub for CSBS students at BMSCE eliminating the chaos of scattered study materials across WhatsApp groups and Google Classroom.
- Built responsive frontend with React.js component-based architecture; integrated Supabase for real-time document storage, user authentication, and secure access control.
- Optimized for high traffic exam week scenarios; platform surfaces curated content without requiring students to dig through disorganized channels.
- Tech Stack: React.js, Supabase (Auth + Real-time DB), Vercel.

RESEARCH

Human-Supervised Dual-UUV System for Autonomous Maritime Surveillance [Research Paper](#) · tusharjain.in/research

Autonomous Systems · Sensor Fusion · Human in the Loop Control · EKF Localization

- Proposed a split-architecture dual-UUV surveillance framework combining autonomous acoustic threat detection (passive UUV) with operator-controlled inspection (active UUV).
- Utilized Extended Kalman Filters (EKF) for precise underwater localization, accounting for detection range limits and underwater acoustic communication constraints.
- Defined a strict Human-in-the-Loop protocol for all active system responses ensuring safety, legal compliance, and ethical operation in maritime security contexts.
- Analyzed theoretical models for localization accuracy, communication range constraints, and coordination latency between dual-vehicle systems.

EDUCATION

Bachelor of Engineering - Computer Science & Business Systems (CSBS)

2024 – 2028

BMS College of Engineering (BMSCE) · Bengaluru, India **CGPA: 9.08 / 10.0** (up to 3rd Semester)

SKILLS & TECHNOLOGIES

Frontend: React.js, Next.js, HTML5, CSS3, JavaScript (ES6+), Vanilla JS, Responsive Design, Mobile-first UI

Backend & Database: Supabase (Auth + Real-time DB), Google Sheets API, Node.js, REST APIs

Hardware & Embedded: ESP32, Arduino, MPU6050 (IMU), MAX30102 (SpO2), BLE/Bluetooth, I2C/UART/SPI, Sensor Fusion, C/C++ Firmware

Programming Languages: JavaScript, Python (Academic), MATLAB (Academic), C, C++

Tools & Platforms: Git/GitHub, Vercel, Razorpay (Test Mode), Android (APK Build & Publish), Antigravity, VS Code

Areas: System Design, Distributed Transactions, Offline-first Architecture, Real-time Sync, Edge Case Management, MVP Strategy

Portfolio: tusharjain.in · All Projects: tusharjain.in/projects · GitHub: github.com/Tusharjain-19 · LinkedIn: in/tushar-jain-781149322